

# Employee Directory Report

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## Working UI Screenshot

The application displays the add/edit form, search and sort controls, seeded employee records, active headcount, and active/inactive status badges.

**HUMAN RESOURCES**

## Employee Directory

Manage staff records with Vue, Axios, Express and MySQL.

7 Total

6 Active

1 Inactive

NEW RECORD

Add Employee

Employee ID

EMP001

Full Name

Aina Sofia Rahman

Email

name@company.my

Department

Select department

Position

Systems Analyst

Hire Date

dd-....-yyyy

Salary (RM)

1500

☒ Active employee

Add Employee

Search

Sort by

Order

Name, ID, email or department

Name

Ascending

Reset

| EMPLOYEE                                                      | DEPARTMENT | POSITION               | HIRE DATE  | SALARY      | STATUS | ACTIONS                           |
|---------------------------------------------------------------|------------|------------------------|------------|-------------|--------|-----------------------------------|
| <b>Aina Sofia Rahman</b><br>EMP001 - aina.rahman@company.my   | HR         | HR Executive           | 2021-03-15 | RM 4,200.00 | Active | <div>Edit</div> <div>Delete</div> |
| <b>Daniel Lim Wei Jian</b><br>EMP002 - daniel.lim@company.my  | IT         | Systems Analyst        | 2020-08-01 | RM 6,800.00 | Active | <div>Edit</div> <div>Delete</div> |
| <b>Farid Iskandar</b><br>EMP006 - farid.iskandar@company.my   | IT         | Frontend Developer     | 2023-06-05 | RM 5,100.00 | Active | <div>Edit</div> <div>Delete</div> |
| <b>Mei Ling Tan</b><br>EMP005 - mei.tan@company.my            | Marketing  | Digital Marketing Lead | 2022-01-17 | RM 6,300.00 | Active | <div>Edit</div> <div>Delete</div> |
| <b>Nur Izzati Hassan</b><br>EMP003 - izzati.hassan@company.my | Finance    | Accountant             | 2019-11-20 | RM 5,900.00 | Active | <div>Edit</div> <div>Delete</div> |

Figure 1: Vue Employee Directory running at <http://127.0.0.1:5174> with MySQL seed data.

## Chapter 8 Learning Outcomes

- **LO1: Connect Vue to a backend.** The Vue app calls the Express REST API on port 3001 and keeps the employee list in the parent component state.
- **LO2: Use Axios.** All HTTP requests go through one Axios instance in `src/services/api.js`, including a request logger and a response error mapper.
- **LO3: Build validated Vue forms.** `EmployeeForm.vue` uses `v-model.trim` and `v-model.number`, validates all required fields, and displays inline errors before any request is sent.
- **LO4: Perform CRUD with MySQL.** Express implements create, read, update, delete, server-side search, and whitelisted sorting with prepared statements through the `mysql2` promise pool.

## Implementation Notes

The application is separated into form, list, and search/sort components. The SQL script creates the database, table, and seven sample records across HR, IT, Finance, Marketing, and Operations, with one inactive employee.

## Challenges and Resolutions

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- **Safe sorting:** Directly accepting a column name from the client would risk SQL injection. The API maps requested sort keys to a fixed whitelist before building the `ORDER BY` clause.
- **Consistent errors:** Network and server errors can have different Axios shapes. The response interceptor converts them into one message format used by the UI banner.
- **Responsive layout:** The form and directory use CSS grid and a horizontally safe table layout so the page remains usable on laptop and tablet widths.